

Analysis group meeting

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17 February, 2025



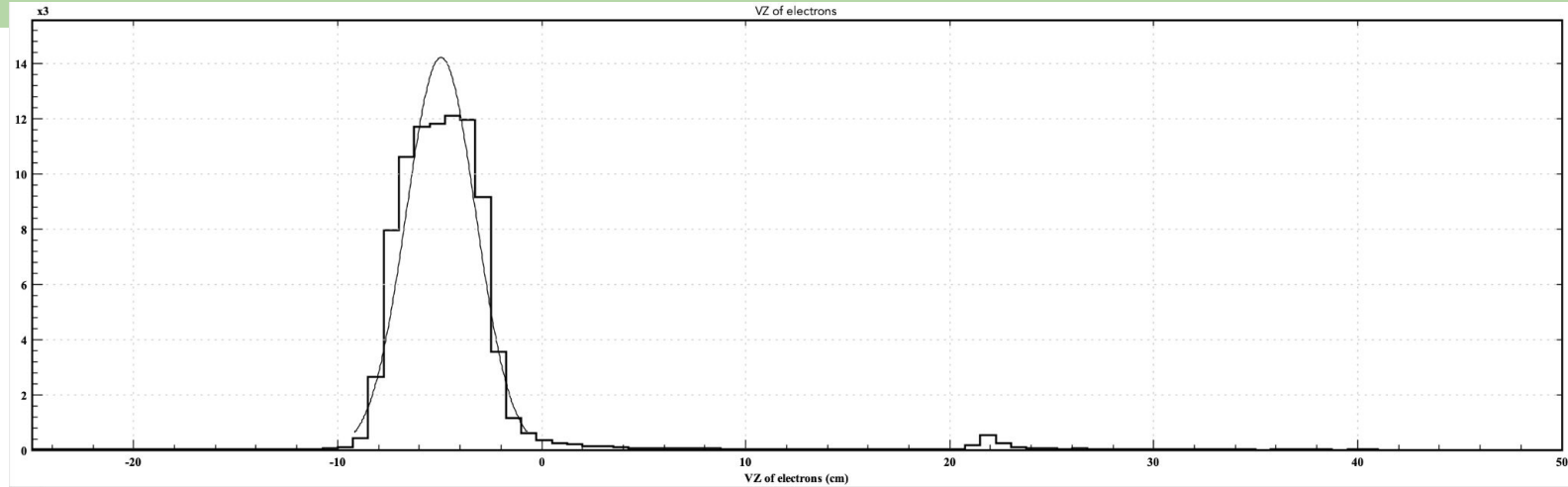
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- Mini-timeline for DC calibration is done and publish
 - https://clas12mon.jlab.org/rgd/pass0v9_dc/tlsummary/
- Cooking for DC calibration for special HV settings runs (18309,18310,18311,18312) is done.
- I have made a pull request for the forward electron/negative/positive Vz which have been accepted by Chris.

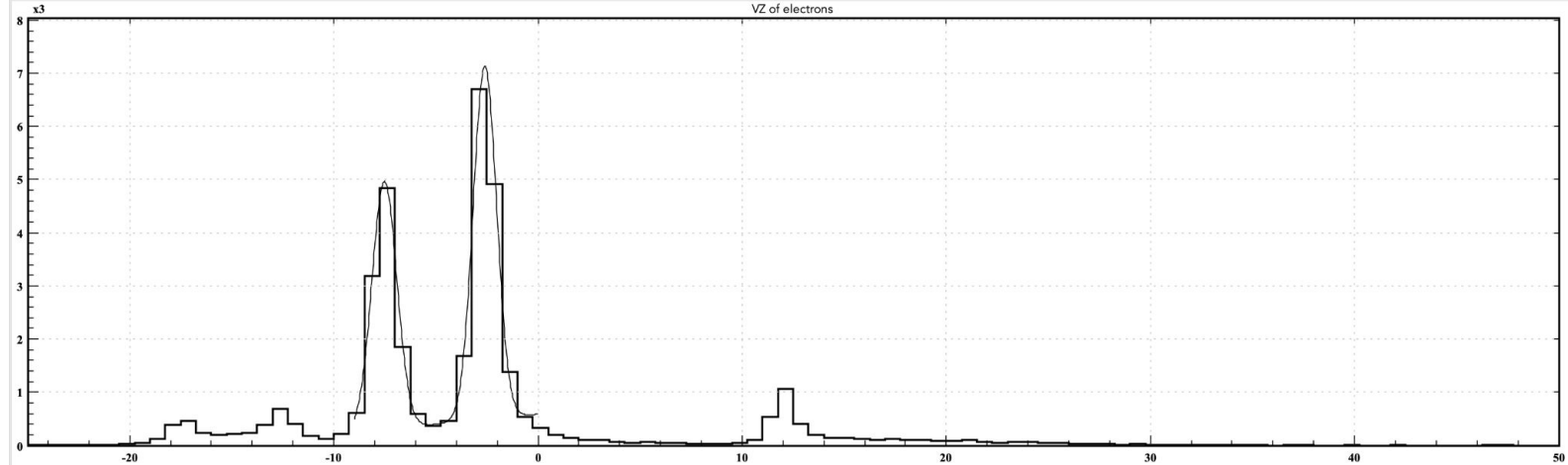
Change range and initial values for the fit function for forward Vz for RG-D. Apply the double gaussian fit only for solid target. #278

 Merged c-dilks merged 3 commits into [JeffersonLab:main](#) from [mathieuouillon:main](#) 3 days ago

Forward Vz for timelines



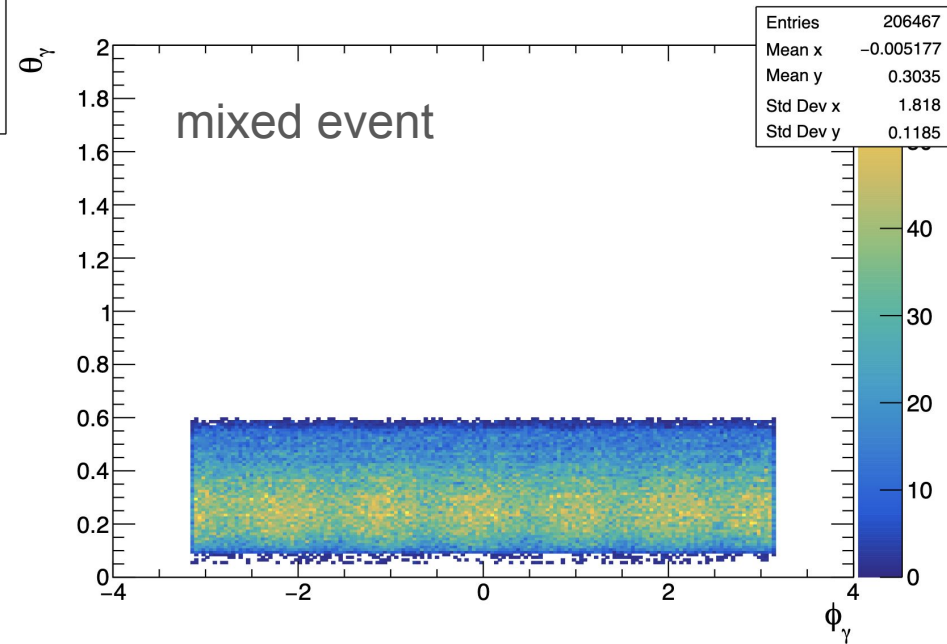
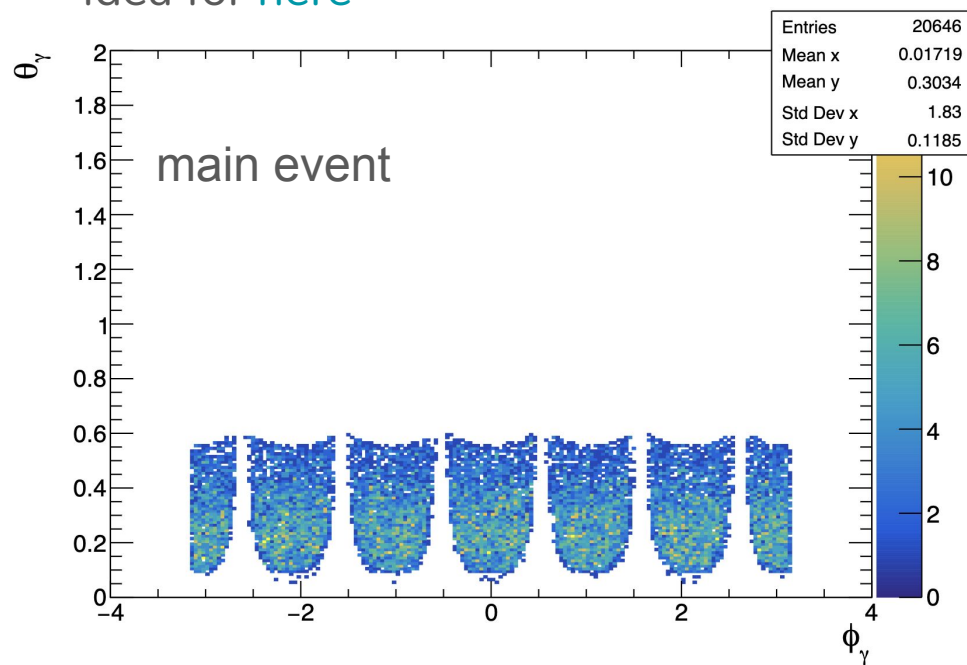
run 18318



run 18356

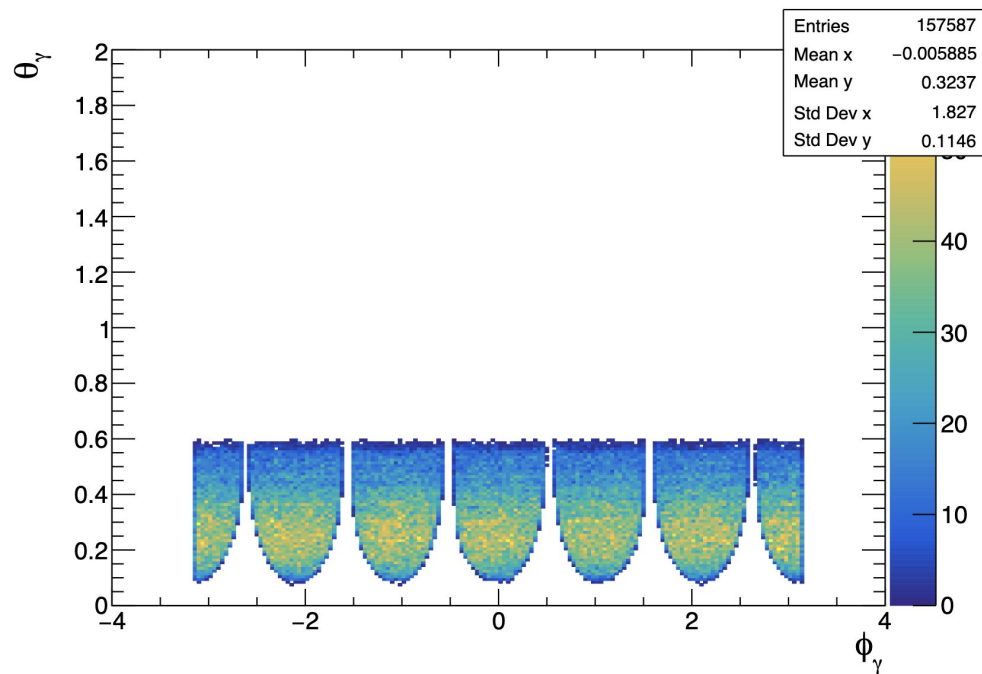
RG-D study: π^0 analysis

- Check why the event mixing method doesn't work.
- Check the phase space for main event and the mixed event.
- Rotate the mixed photon by $\Delta\varphi = \varphi_{e,\text{main}} - \varphi_{e,\text{mixed}}$
- Idea for [here](#)

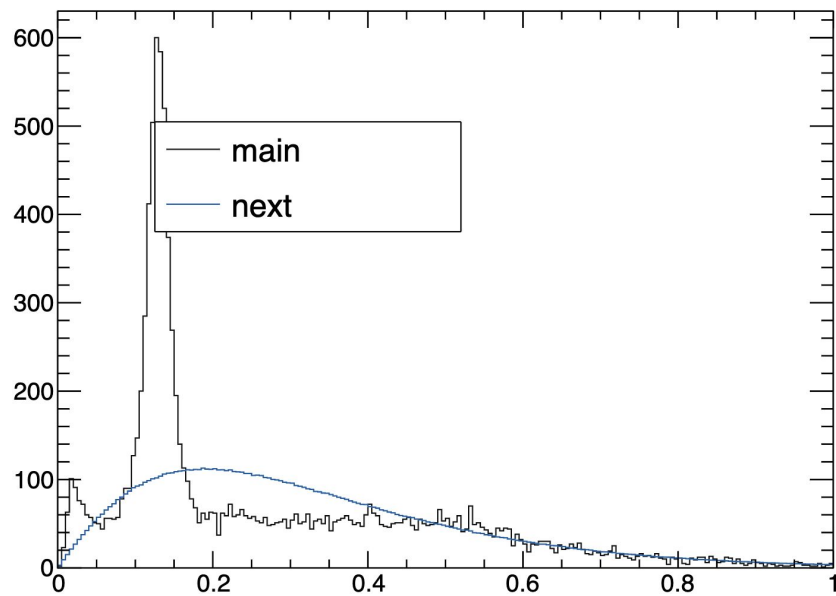


RG-D study: π^0 analysis

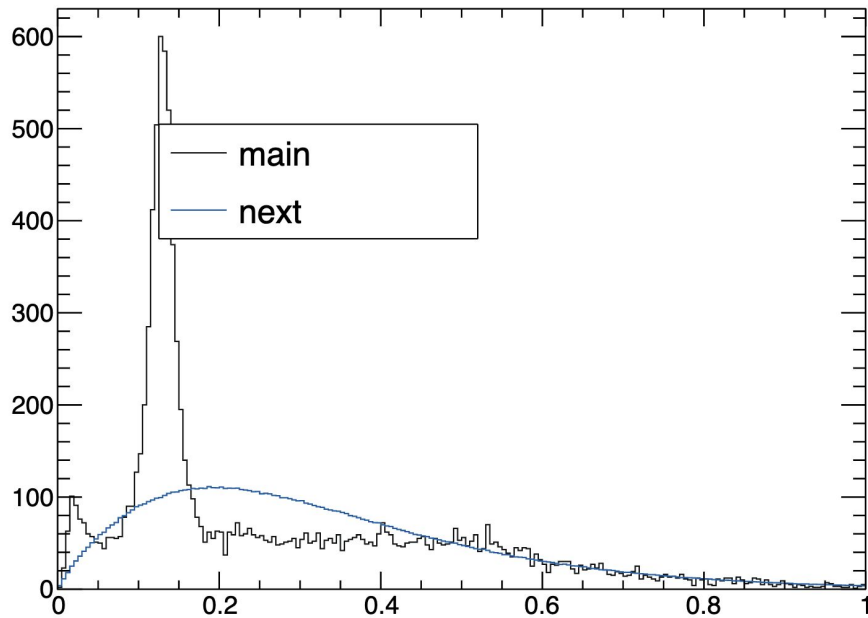
- Apply “fiducial” cut on the mixed photons



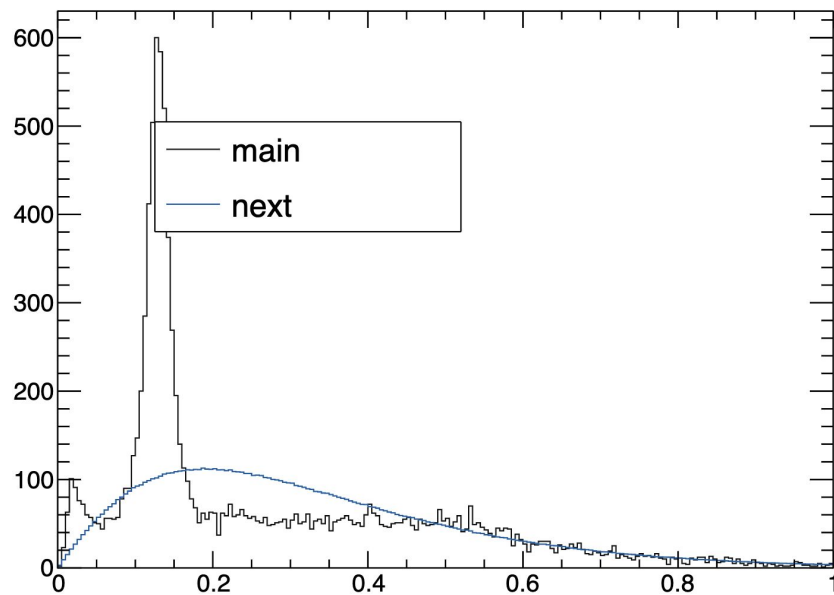
without “fiducial” cut



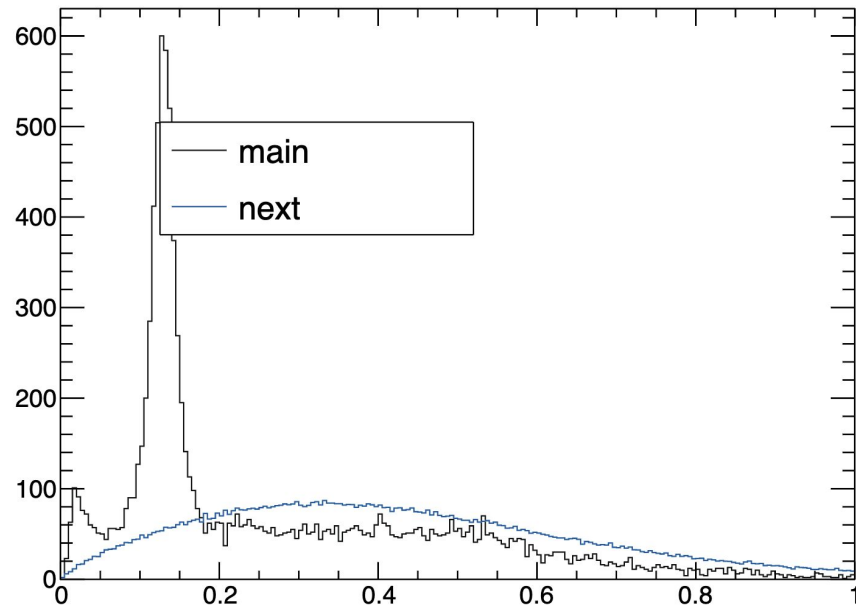
with “fiducial” cut



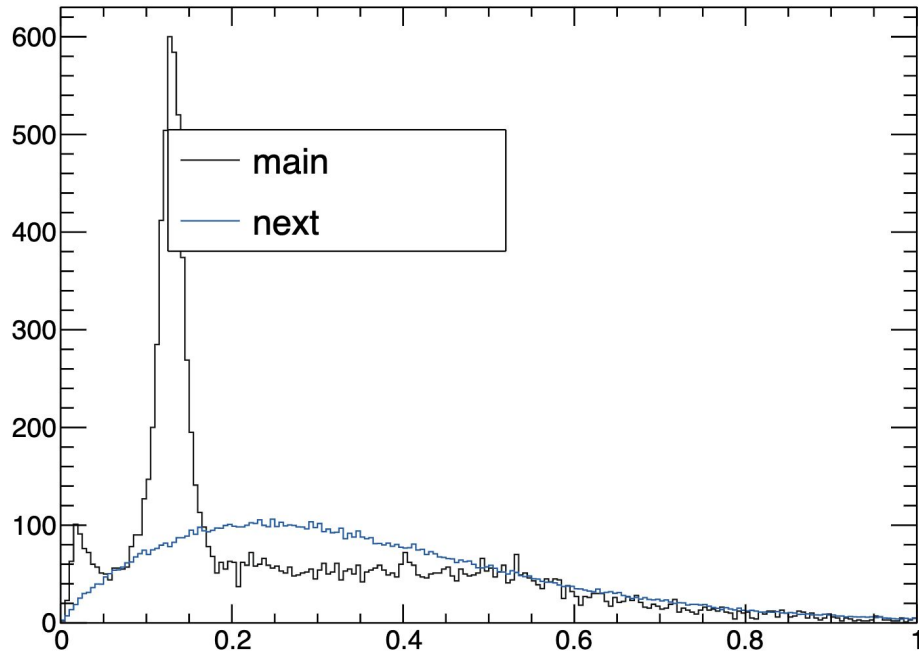
without “fiducial” cut



without rotation



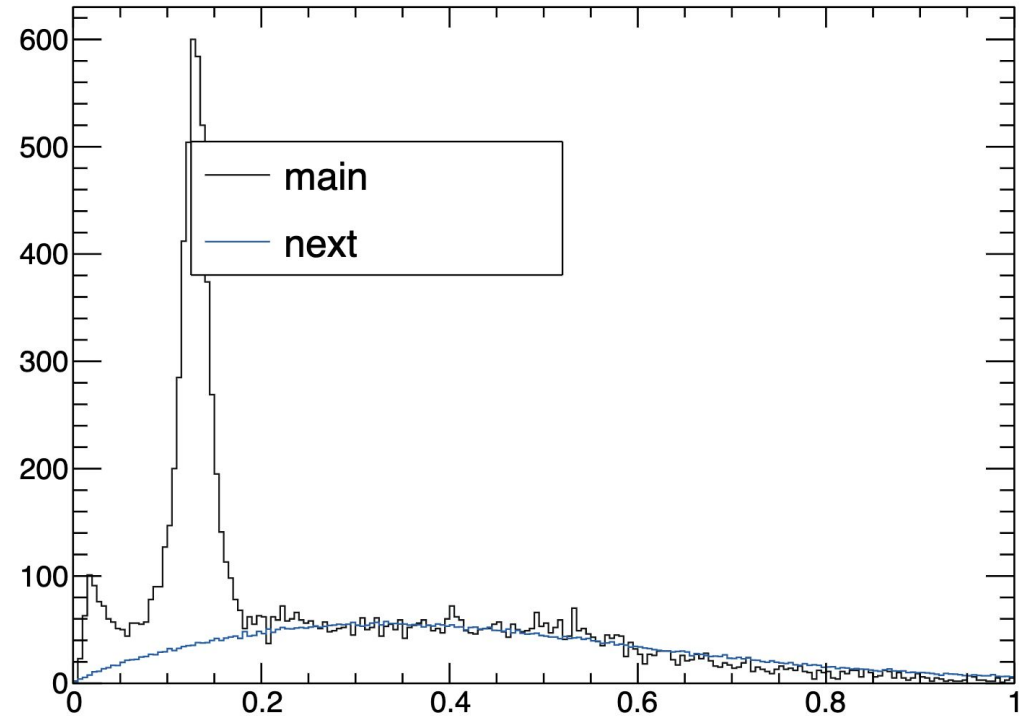
- Add cut: $0.3 < z_h < 0.7$ for the mixed event and $\theta_{ey} > 8$ deg



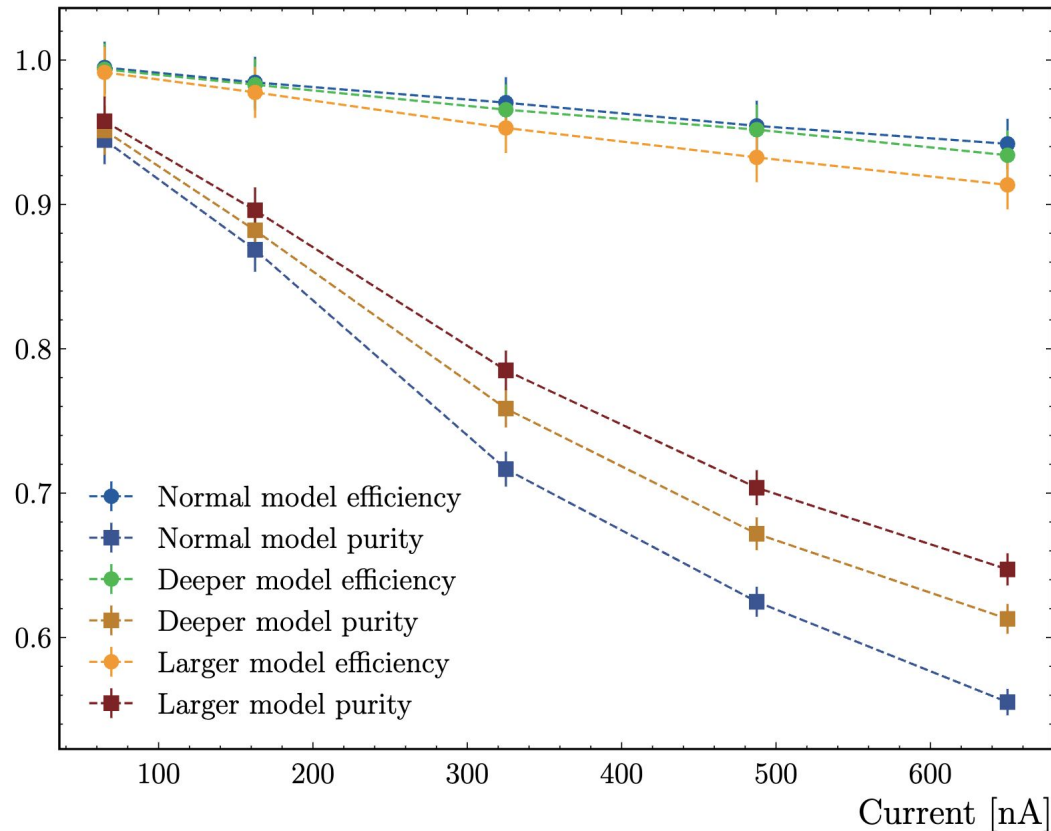
Without the rotation.

With cuts on zh and opening angle.

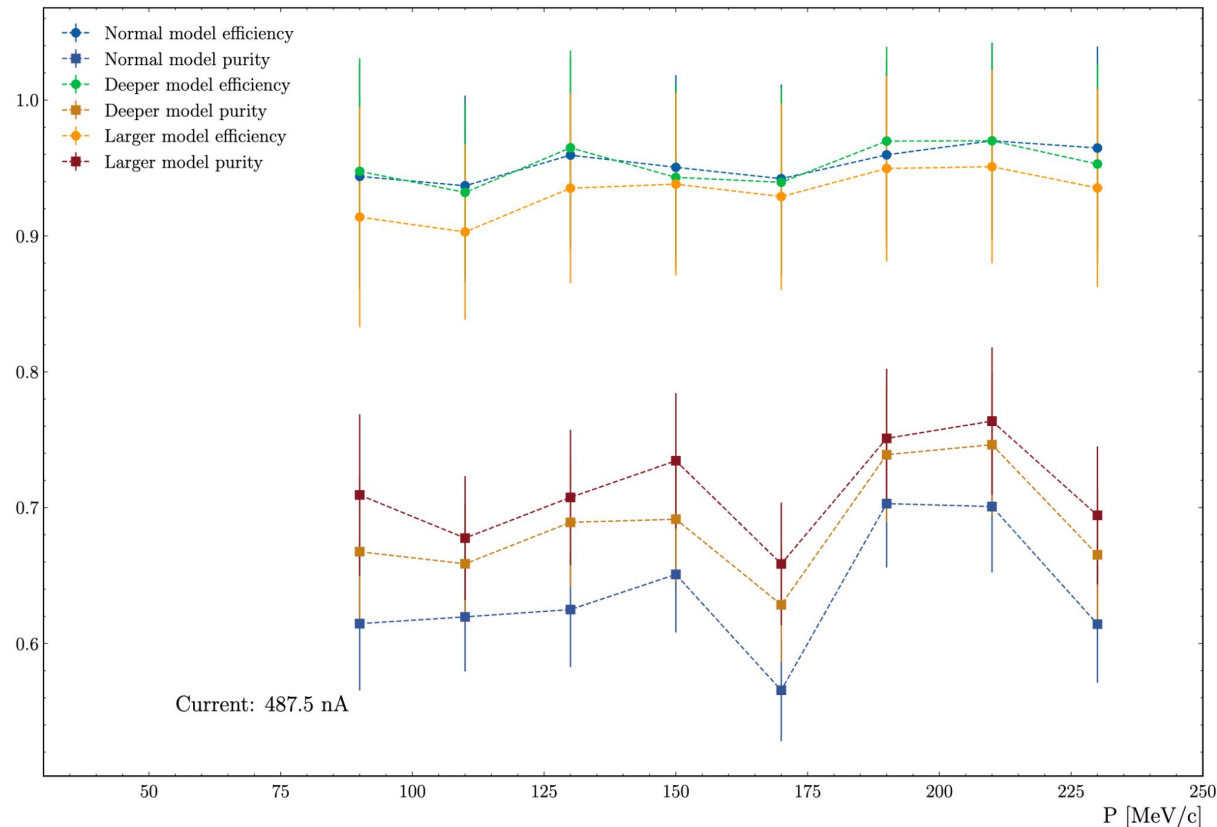
If I choose the scale factor.



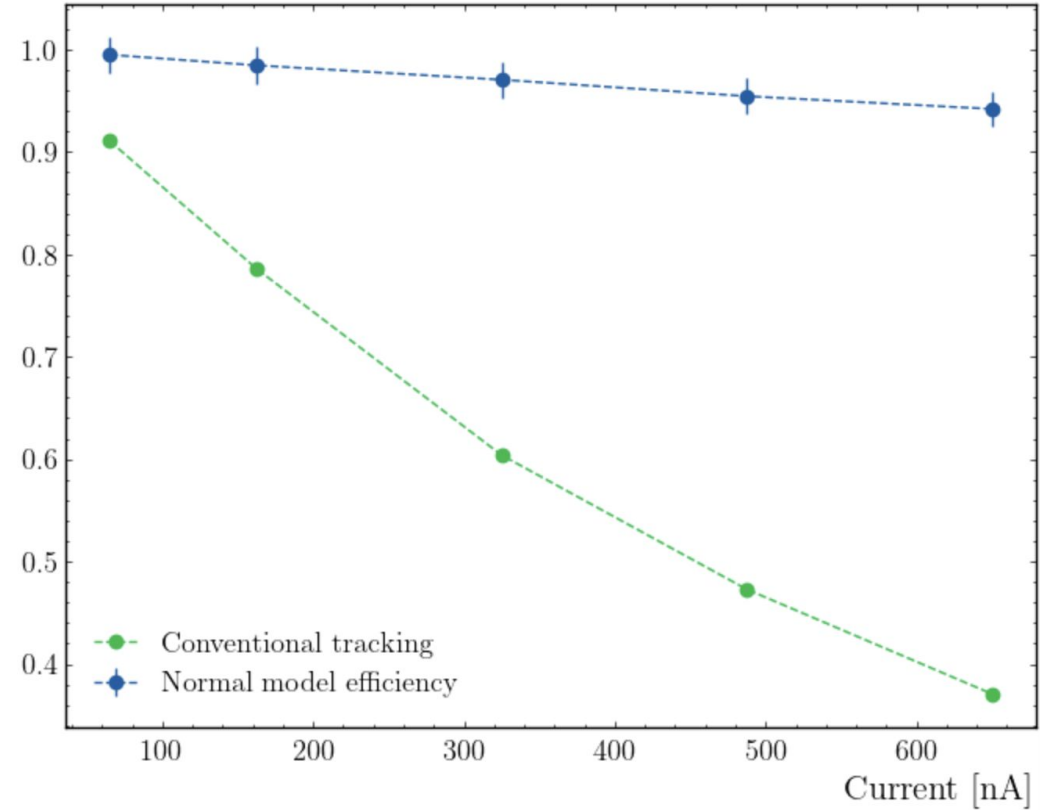
- Use simulated background (mostly electron-electron collision)
- Add proton track.
- Compute efficiency and purity:
 - Efficiency: Number of good tracks classified as good / number of event which have a track.
 - Purity: Number of good tracks classified as good / number of tracks (good or bad) classified as good.



- Use simulated background (mostly electron-electron collision)
- Add proton track.
- Can also compute the efficiency as function of the proton momentum



- We can also compare the efficiency for the conventional track finding and the AI.



- Work on the presentations, both the AI for ALERT and RG-D update.
- RG-D analysis:
 - Cross check for the nuclear transparency vs Q^2 , find why the nuclear transparency for Cu and Sn is so high.
 - Work on π^0 analysis: Fix the event mixing method
- Background merging, generate 100k events for one configuration